KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Multiple variables - mixed practice

Mix two variables, three variables, squares and products.

10 MINUTES / 6 QUESTIONS / AS-10

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $x = 6, y = 5$

Evaluate $6x + 5y$

2 Given $a = -2, b = -1, c = 0$

Evaluate $4a - 4b + 3c$

3 Given $a = 2, b = -3$

Evaluate $a^2 - 6ab + b^2$

4 Given $x = 2, y = 1$

Evaluate $5x + 2y$

5 Given $a = -3, b = -4, c = 1$

Evaluate $2a - 6b + 2c$

6 Given $a = -3, b = 1$

Evaluate $a^2 - 6ab + b^2$

Answer sheet

Multiple variables - mixed practice | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 61

02 -4

03 49

04 12

05 20

06 28

Worked solutions

Multiple variables - mixed practice

01 Given $x = 6, y = 5$

$6x + 5y$

Substitute $x = 6, y = 5$.

$6\times 6 + 5\times 5$

$36 + 25$

61

02 Given $a = -2, b = -1, c = 0$

$4a - 4b + 3c$

Substitute values.

$4\times(-2) - 4\times(-1) + 3\times 0$

$-8 + 4$

-4

03 Given $a = 2, b = -3$

$a^2 - 6ab + b^2$

Substitute $a = 2, b = -3$.

$(2)^2 - 6(2)(-3) + (-3)^2$

$4 + 36 + 9$

49

04 Given $x = 2, y = 1$

$5x + 2y$

Substitute $x = 2, y = 1$.

$5\times 2 + 2\times 1$

$10 + 2$

12

Worked solutions

Multiple variables - mixed practice

05 Given $a = -3$, $b = -4$, $c = 1$

$2a - 6b + 2c$

Substitute values.

$2\times(-3) - 6\times(-4) + 2\times 1$

$-6 + 24 + 2$

20

06 Given $a = -3$, $b = 1$

$a^2 - 6ab + b^2$

Substitute $a = -3$, $b = 1$.

$(-3)^2 - 6(-3)(1) + (1)^2$

$9 + 18 + 1$

28